DO NOW — RECALL

Timed Paper 2 algorithms practice

Name: _____ Class: _____ Date: _____

1. Which structure retrieves the most recently added item first?

2. What is the sorted-input precondition?

3. What moves right as a key is inserted into a sorted prefix?

4. What are the worst-case time complexities of linear and binary searches?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Stack (LIFO).

2. Model answer

The array must be sorted.

3. Model answer

All preceding values greater than the key.

4. Model answer

Linear O(n); binary O(log n) for sorted input.

Vocabulary — know these terms

trace — Record an algorithm's changing state.

sorted prefix — Part already in order during insertion sort.

precondition — Requirement, such as ordering, for an algorithm.

O(log n) — Work grows by repeated halving.

auxiliary space — Extra storage required by an algorithm.

NEW LEARNING

Read, annotate and explain

Read the task command

A complete pass means all permitted adjacent comparisons for that pass, not just one swap.

Trace one pass

[6,2,5,1] becomes [2,5,1,6] after comparing each pair.

Finish every pass

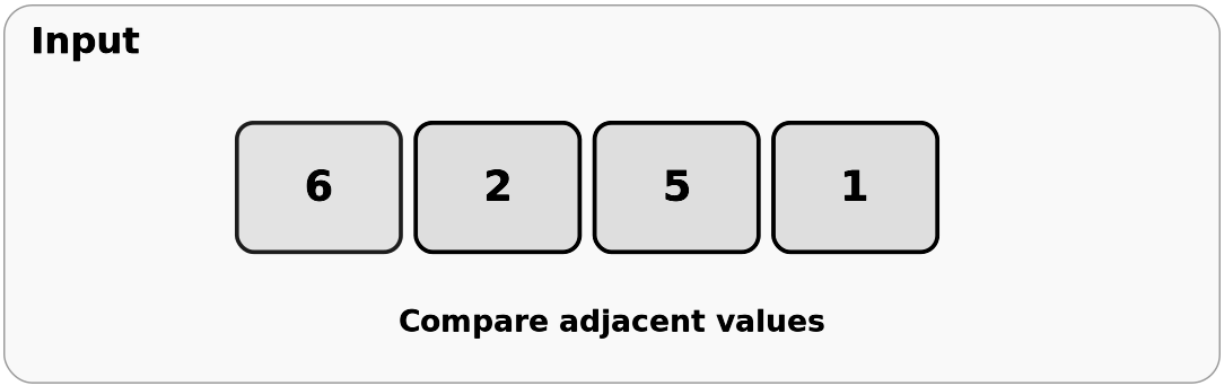
Continue to [2,1,5,6], then [1,2,5,6]. A clear trace shows each pass.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

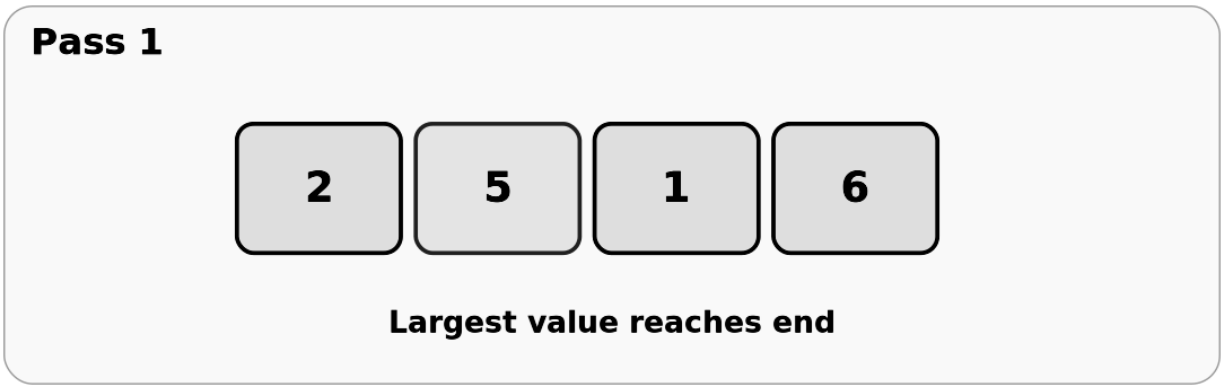
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Input 6, 2, 5, 1 Compare adjacent values

New Learning

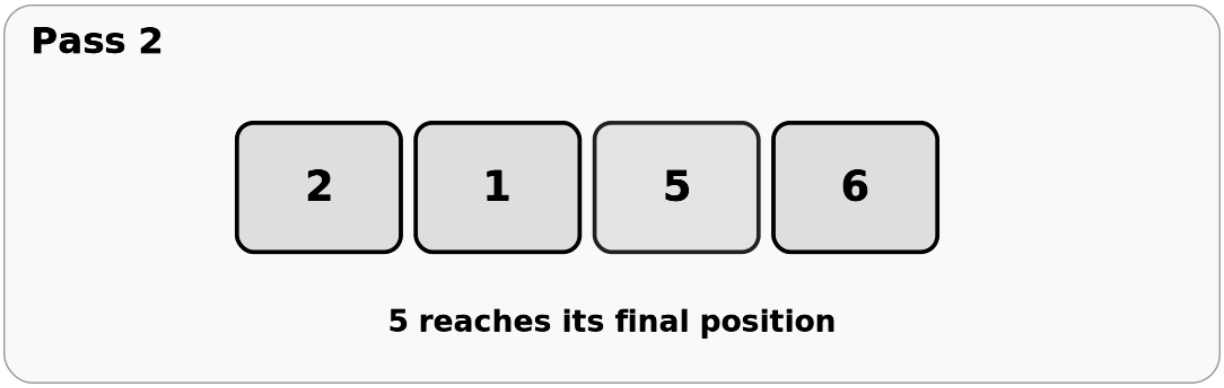


Pass 1 2, 5, 1, 6 Largest value reaches end

New Learning · original lesson diagrams

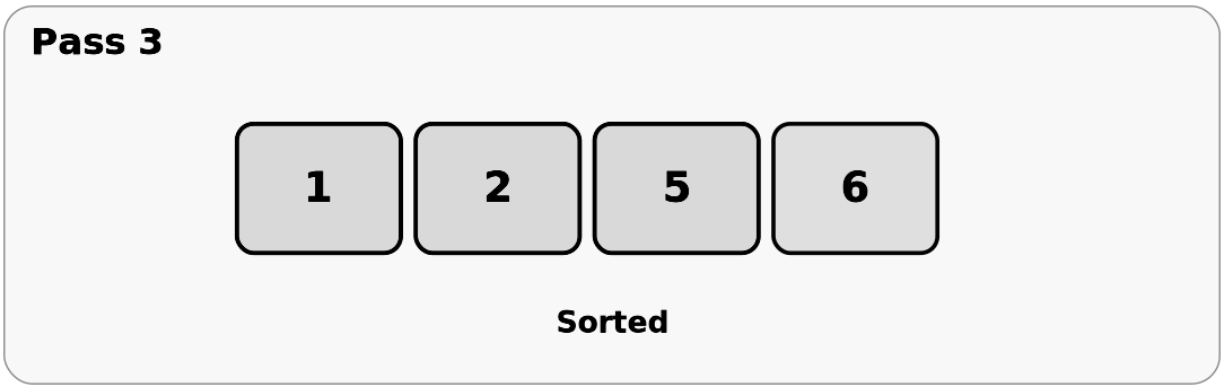
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Pass 2 2, 1, 5, 6 5 reaches its final position

New Learning



Pass 3 1, 2, 5, 6 Sorted

I DO — WORKED EXAMPLE

Follow the worked method

Model the first complete ascending bubble-sort pass on [4,1,3].

Predict first

Start [4,1,3] and compare 4 with 1: swap.

Check the method

Now compare 4 with 3: swap, producing [1,3,4].

Check the method

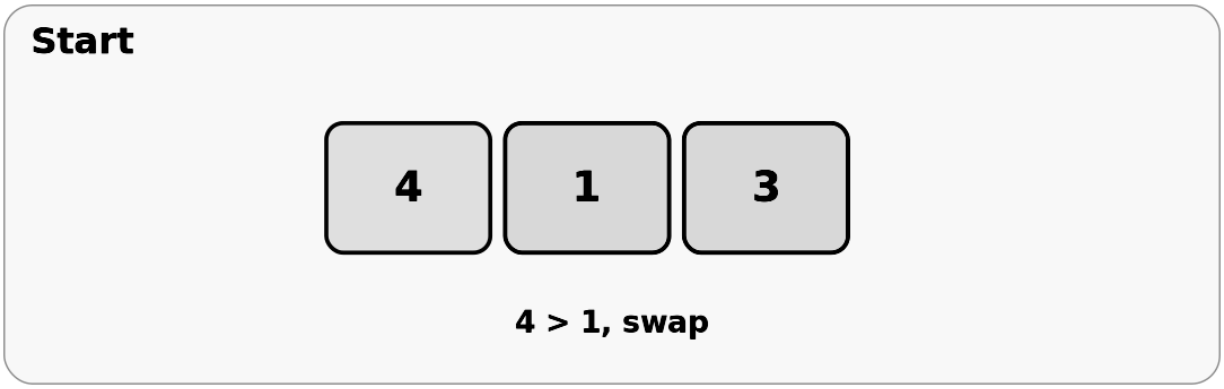
The pass is complete and the array is sorted.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Model the first complete ascending bubble-sort pass on [4,1,3].

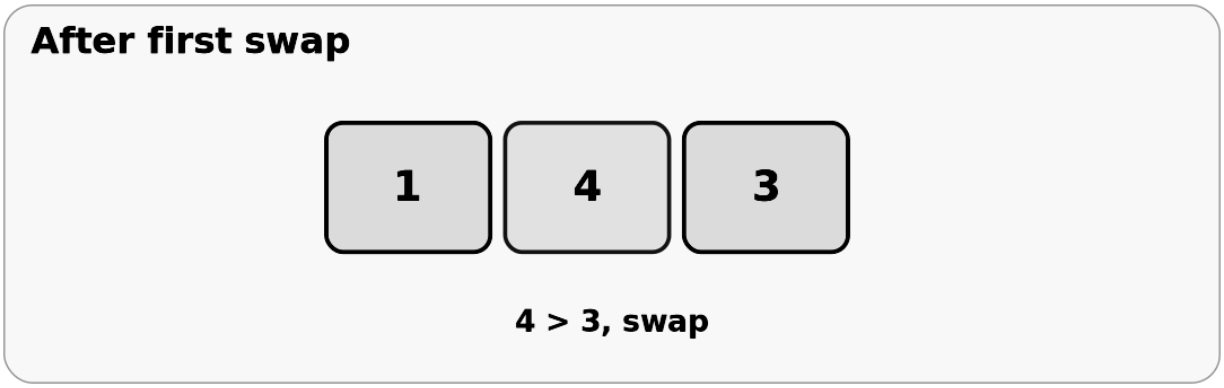
Start [4,1,3] and compare 4 with 1: swap.



Start 4, 1, 3 4 > 1, swap

Model the first complete ascending bubble-sort pass on [4,1,3].

Start [4,1,3] and compare 4 with 1: swap.



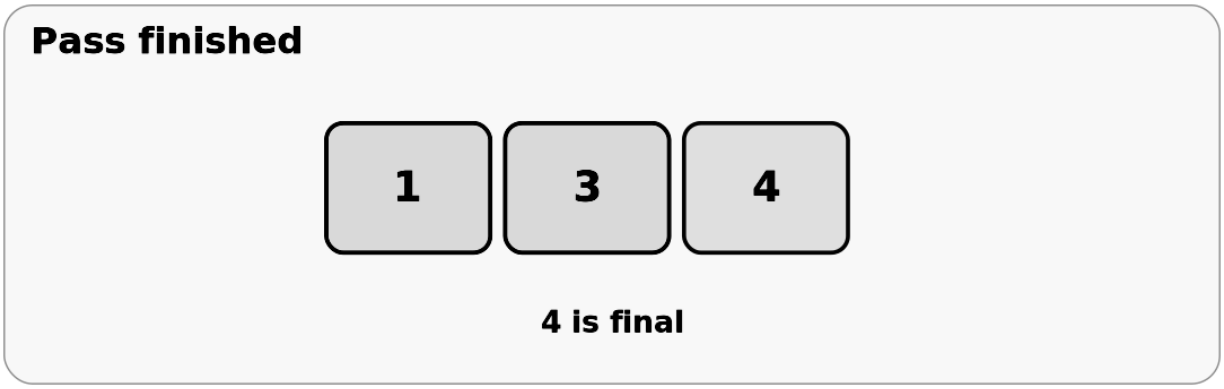
After first swap 1, 4, 3 4 > 3, swap

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Model the first complete ascending bubble-sort pass on [4,1,3].

Start [4,1,3] and compare 4 with 1: swap.



Pass finished 1, 3, 4 4 is final

WE DO — GUIDED PRACTICE

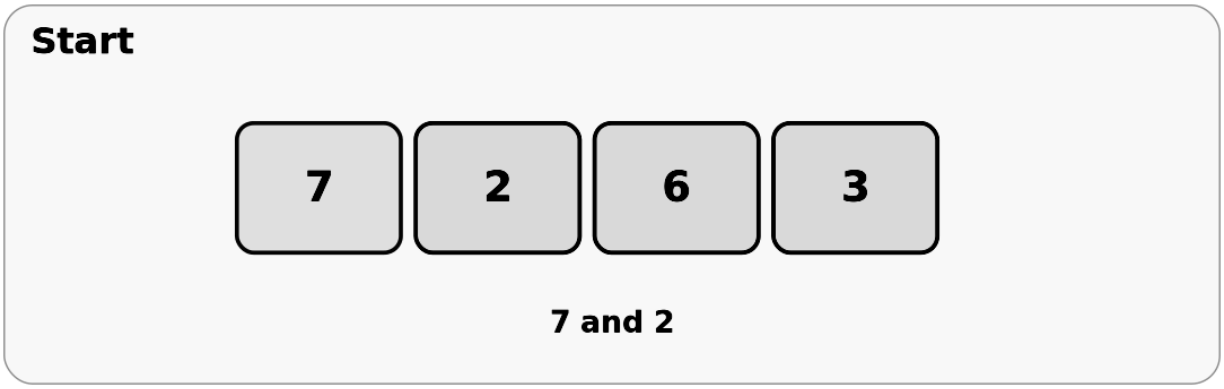
Try → Discuss → Improve

We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Trace one full bubble-sort pass on [7,2,6,3].

7 swaps with 2 giving [2,7,6,3]; then 7 swaps with 6 giving [2,6,7,3].



Start 7, 2, 6, 3 7 and 2

WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

Guided 1: Predict first 7 swaps with 2 giving [2,7,6,3]; then 7 swaps with 6 giving [2,6,7,3].

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

IMPROVE — Write one precise correction or stronger explanation in your own answer.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

An array contains [8,3,6,1]. Trace ascending insertion sort, giving the array after each outer-loop iteration and explaining how the key is positioned.

Question 2

Write a C# function BinarySearch(int[] a,int target) returning the first matching index it encounters, or −1. Include correct bounds and a sorted-input precondition.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Explain one advantage and one disadvantage of merge sort over bubble sort, including time complexity, and identify an input scenario for which linear search is more practical than binary search.

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

Binary search works correctly on any unordered input.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

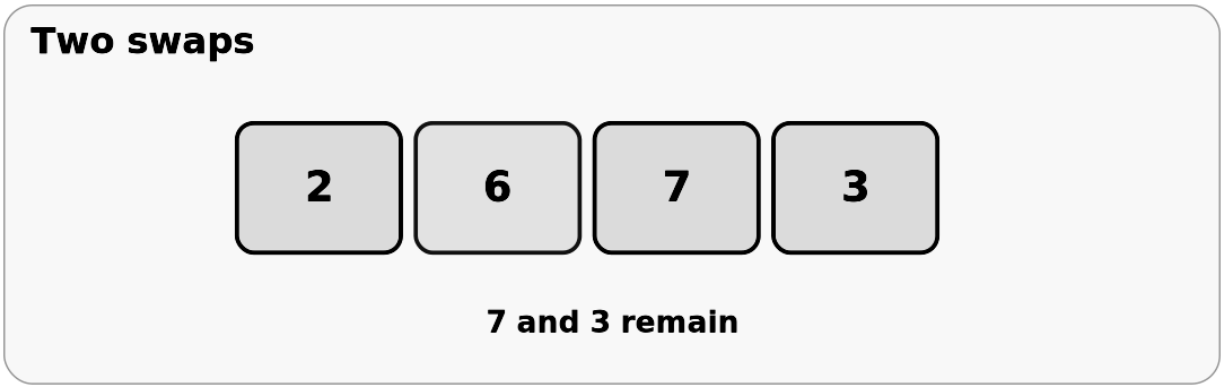
Next step: revisit the model or a relevant revision-guide section if needed.

We Do Check · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Trace one full bubble-sort pass on [7,2,6,3].

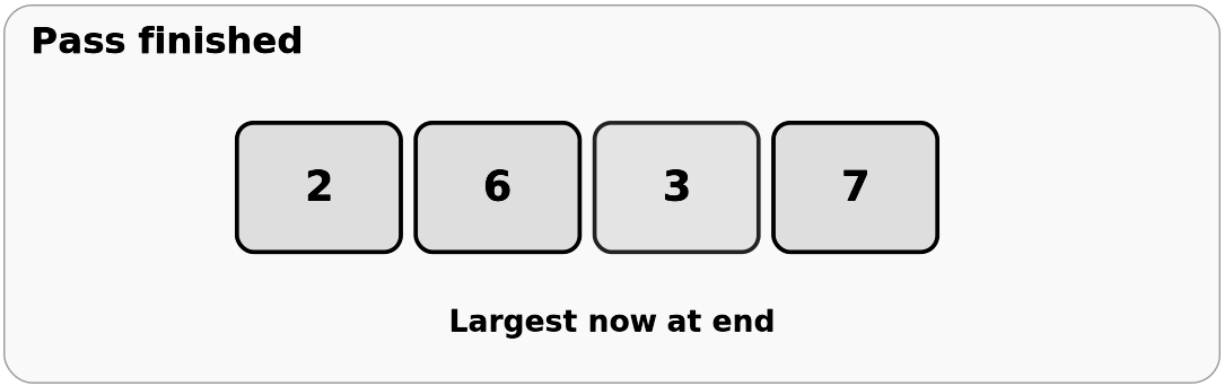
7 swaps with 2 giving [2,7,6,3]; then 7 swaps with 6 giving [2,6,7,3].



Two swaps 2, 6, 7, 3 7 and 3 remain

Trace one full bubble-sort pass on [7,2,6,3].

7 swaps with 2 giving [2,7,6,3]; then 7 swaps with 6 giving [2,6,7,3].



Pass finished 2, 6, 3, 7 Largest now at end

WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

Guided 1 — original CHECK

Check the method

Compare 7 with 3: swap to [2,6,3,7].

The complete first pass ends [2,6,3,7], not yet sorted.

IMPROVE: Explain one correction to your own reasoning.